DEPARTMENT OF
COMPUTER SCIENCE

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BSc in ⟨Computer Science⟩

PRIVACY-ENHANCED ONTOLOGIES

TODO: CHOOSE SUBTITLE

MASTER IN COMPUTER SCIENCE

NOVA University Lisbon
Draft: December 20, 2021

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Abstract

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution?
4. What results (implications/consequences) from the solution?

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4

Resumo

1. Qual é o problema?
2. Porque é que é um problema interessante/desafiante?
3. Qual é a proposta de abordagem/solução?
4. Quais são as consequências/resultados da solução proposta?

Palavras-chave: Palavra-chave 1, Palavra-chave 2, Palavra-chave 3, ...

Contents

Todo list

1

Introduction

1.1 Context

1. Briefly contextualize the Semantic Web. Why are ontologies relevant? T[1]
2. State of adoption. How/where are ontologies being used? T[2]
3. Future expectations. How will the technology evolve? T[3]

1) *TODO:* Explain the motivation for the Semantic Web and use of ontologies.

2) *TODO:* Enumerate industry/investigation examples.

3) *TODO:* Mention cloud and external computing, general adoption growth...

In 2001, the term *Semantic Web*[1] was first used to define what would be the evolution of the *World Wide Web*. The *World Wide Web* solved the challenge of distributing information at scale and globally, however it was designed for human use. In the traditional *Web* documents are easily accessible and readable by humans, but for automated agents little more than document retrieval is possible. Contrary to humans, machines can not reason over the information they access if they do not know the semantics of the data.

The *Semantic Web* addresses this issue, by redesigning the representation of information, in a way that not only preserves meaning and relationships but is also machine-readable. A concept is mapped to a unique identifier, and meaning is encoded in sets of triples, each triple consisting of a subject, verb and object (similar to an elementary sentence). [1] To represent relevant concepts, and their relations, in specific domains, *ontologies* are then built, using these specifications.

1) **[Diogo]:** Very general explanation because details will be covered in the State-of-the-Art chapter.

This novel method of knowledge representation differs from the previous, centralized way, and greatly simplifies the process of interconnecting different *knowledge bases*. With centralized representations both parties need to agree on the semantics of the data, in contrast, using ontologies, information is semantically linked by default. In the *Semantic Web*, *automated agents* can truly communicate, share and reason about distinct *knowledge bases*.

1.2 Problem

1. How is the current state of ontology development/engineering? T[4]

4) *TODO:* Explain tools and workflow used in ontology engineering.

2. Why remote computation is gaining adoption in the area, but what are current limitations/problems unresolved? T[5]

3. Why is there need for privacy in ontologies? T[6]

5) *TODO:* Contrast the usefulness with the limitation/security problems of cloud/remote computing

6) *TODO:* Explain why some ontologies need privacy mechanisms. Define the mechanisms needed.

1.3 Proposed Solution

Briefly explain proposed solution. The scope of the solution, contributions, and how will it solve the problems mentioned in previous section.

1.4 Document Organization

Specify the document organization, per chapter.

2

State of the Art

3

Work Plan

Bibliography

[1] T. Berners-Lee, J. Hendler, and O. Lassila. "The semantic web". In: *Scientific american* 284.5 (2001), pp. 34–43 (cit. on p. 1).